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'LTE-diffusion approximation' for arc calculations

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Abstract

This paper proposes the use of the 'LTE-diffusion approximation' for predicting the properties of electric arcs. Under this approximation, local thermodynamic equilibrium (LTE) is assumed, with a particular mesh size near the electrodes chosen to be equal to the 'diffusion length', based on D_e/W, where D_e is the electron diffusion coefficient and W is the electron drift velocity. This approximation overcomes the problem that the equilibrium electrical conductivity in the arc near the electrodes is almost zero, which makes accurate calculations using LTE impossible in the limit of small mesh size, as then voltages would tend towards infinity. Use of the LTE-diffusion approximation for a 200 A arc with a thermionic cathode gives predictions of total arc voltage, electrode temperatures, arc temperatures and radial profiles of heat flux density and current density at the anode that are in approximate agreement with more accurate calculations which include an account of the diffusion of electric charges to the electrodes, and also with experimental results. Calculations, which include diffusion of charges, agree with experimental results of current and heat flux density as a function of radius if the Milne boundary condition is used at the anode surface rather than imposing zero charge density at the anode.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Although many advances have been made in recent years in the modelling of electric arc processes, improvements in modelling capabilities are still needed to make possible rapid calculations of the time dependence of arc properties in three dimensions, as are required, for example, in arc welding.

In previous papers (Tanaka *et al* 2003, Ushio *et al* 2004, Lowke *et al* 2005) we have presented calculations of temperatures of free burning arcs with tungsten cathodes for currents of the order of 100–200 A with an account of non-equilibrium electron densities throughout the arc due to the influence of ambipolar diffusion. Inclusion of diffusion currents at the electrodes overcomes the problem of the equilibrium electrical conductivity at the electrodes being almost zero as the equilibrium electron density at the electrode temperatures is almost zero. The predictions of both arc and electrode temperatures are in good agreement with experimental results.

In the present paper we discuss the use of the local thermodynamic equilibrium (LTE) approximation, but using a particular mesh size at the electrodes equal to the diffusion length of electrons in the arc. We call this approximation 'the LTE-diffusion approximation'. It is found that the approximation, with a simple approximate diffusion length independent of arc radius at the electrode, allows the overall arc voltage, and thus heat energy to the anode, to be calculated reasonably accurately. Then, reasonably accurate calculations can be made of, for example, electrode melting, which is particularly important for properties of arc welding. Using the 'LTE diffusion approximation' computation time is greatly reduced because the need to calculate effects of the non-equilibrium electron density is avoided.

The physical model for calculating the non-equilibrium effects of electron density, initially presented by Sansonnens *et al* (2000), is reviewed in section 2. This treatment neglects space charge effects at the electrodes where the electron density is not equal to the positive ion density. An earlier

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investigation by Lowke and Quartel (1997) indicates that such non-equilibrium space charge effects at the anode and a thermionic cathode occur within 2 μm. At the anode these effects contribute less than 0.2 V to the total voltage. For thermionic cathodes these non-equilibrium space charge effects are also small for currents of 100–200 A, contributing about 1–2 V to the total arc voltage of about 10 V for 5 mm long arcs. This voltage is necessary to accelerate the electrons from the energy corresponding to thermionic temperatures to the energy corresponding to temperatures of the arc of about 20 000 K. Space charge effects are considered in the papers of Benilov, e.g. Benilov and Marotta (1995). There are also non-equilibrium effects at the electrodes due to the temperatures of the electrons and heavy particles not being equal; (Jenista and Heberlein 1997, Tanaka and Ushio 1999, Tanaka *et al* 1999, Liau *et al* 2003). However, effects of non-equilibrium of charge densities, where the electron density can vary by orders of magnitude from the equilibrium value, are generally much more important than effects of non-equilibrium temperatures of different particles, as charge densities influence the resistivity of the plasma near the electrodes and thus affect plasma heating. Differences in temperature between the electrons and heavy particles certainly occur, but are expected to influence the means of conduction rather than the total conduction energy to the anode. Our numerical scheme conserves total energy.

In section 3 the methodology and physical basis of the 'LTE-diffusion approximation' is presented, under which we assume LTE, but use a mesh size at the electrodes set at the diffusion length, Λ, defined by D_e/W, where D_e is the electron diffusion coefficient and W is the electron drift velocity. This diffusion length, which is a property of the arc plasma, will be shown to be ~0.1 mm for arcs with currents of the order of 200 A at 1 bar, and is distinct from some other definitions of diffusion length, for example, defined by $1/\Lambda^2 = -\nabla^2 n_e/n_e$, in which case Λ is just proportional to the radius of the discharge, at least for ionization linear in n_e (Rogoff 1981). Section 4 discusses the effect of neglect in our approximation of the large diffusion currents that occur at the electrodes. Section 5 gives a discussion of heat transfer effects at the anode surface.

Predictions including electron diffusion require a boundary condition for electron density at the anode surface. Our previous predictions have used the conventional boundary condition of zero electron density at the anode surface (Huxley and Crompton 1974). But the physical boundary condition that we need to simulate is that electrons moving towards the anode are absorbed, but no electrons travel back from the anode to the plasma, at least for the case of zero reflection of electrons at the anode. An analogous boundary condition has been studied extensively in astrophysics where at the surface of a star or the sun, photons leave to outer space, but a negligible number of photons come from outer space back to the star. From these astrophysical studies, it has been shown that the appropriate boundary condition is to set the gradient of the photon density at the sun's surface so that the extrapolated density is zero at about one mean free path beyond the boundary. This is the so-called Milne boundary condition, first formulated by the astrophysicist Milne (1930) and also considered in general transport theory (Kourganoff 1952, Lindenfeld and Shizgal 1983, Williams 1971). An approximate derivation of this same boundary condition for electrons at an anode has been obtained from the Boltzmann equation (Lowke *et al* 1977). Predictions obtained using this boundary condition are discussed in section 6. It is found that only the Milne boundary condition gives good agreement with experimental results of current density as a function of radius, whereas the condition of zero electron density actually leads to divergent results if very small mesh sizes are used with our ambipolar diffusion formulation.

In section 7 we give a comparison with experimental results of predictions of heat flux and current flux density as a function of radius with predictions from the LTE-diffusion approximation and calculations including electron diffusion with the different boundary conditions.

2. Non-equilibrium of charge density

Non-equilibrium of the densities of charged particles of the plasma adjacent to the electrodes occurs because of the diffusion resulting from the very large gradients in charge density in these regions. The anode is an almost perfect absorber of electrons, so that charge densities near the anode surface do not correspond to the local temperature equilibrium values. At the cathode, positive ions may be easily absorbed and there are further effects due to the emission of electrons from the hot thermionically emitting tungsten.

Near the electrodes charge densities are determined by ambipolar diffusion of electrons from the plasma to the electrodes. The electron continuity equation is solved for the arc region to determine these non-equilibrium electron and ion densities, assumed to be equal. The electron continuity equation in terms of the ambipolar diffusion coefficient D_a is (Morrow and Lowke 1993)

$$\nabla \cdot (D_a \nabla \cdot n_e) + \gamma (G n_e n_0 - n_e^3) = 0, \qquad (1)$$

where $D_a = (D_e/\mu_e + D_i/\mu_i)/(1/\mu_e + 1/\mu_i)$ is the ambipolar diffusion coefficient, n_e is the electron density, γ is the three-body recombination coefficient, $G = n_e^{*2}/n_0^*$ is the Saha function, n_e is the electron density, n_0 is the neutral particle density, n_e^* and n_0^* are densities at LTE and μ_e and μ_i are the electron and ion mobilities. The function G, which is a function of temperature, ensures that in the absence of gradients, the solution for n_e from equation (1) is the equilibrium value of n_e, i.e. n_e^*. The recombination coefficient is also required as a function of temperature. The ratios of diffusion to mobility for electrons and ions, i.e. D_e/μ_e and D_i/μ_i are approximately kT/e, where k is Boltzmann's constant, e the electronic charge and T the local temperature. Electron and ion temperatures are assumed to be the same, even in the electrode regions. The values of mobility for ions and electrons are required as a function of temperature. This treatment assumes that effects from the non-equilibrium of charge densities, which can differ by orders of magnitude from LTE values, are of more significance than non-LTE temperatures. Electrons may retain high temperatures from the arc column near the anode, but total heat transfer to the anode is not likely to be grossly affected, the only difference being whether the transfer is by electron impact with the anode or by thermal conduction.

Net production or destruction of particles. This is zero in case of equilibrium.

Also note, Da_gradne is not the flux of electrons, because, E filed not imposes the condition of je=ji

Equation (1) follows from the two conservation equations for the electron and ion density, i.e.

$$\nabla \cdot \boldsymbol{j}_e = \gamma(G n_e n_0 - n_e^3), \tag{2}$$

$$\nabla \cdot \boldsymbol{j}_i = \gamma(G n_e n_0 - n_e^3), \tag{3}$$

and also the two expressions for electron and ion flux density in terms of drift and diffusion, i.e.

$$\boldsymbol{j}_e = n_e \mu_e \boldsymbol{E} - D_e \nabla n_e, \tag{4}$$

$$\boldsymbol{j}_i = -n_i \mu_i \boldsymbol{E} - D_i \nabla n_i, \tag{5}$$

where $\boldsymbol{j}_e$ and $\boldsymbol{j}_i$ are the electron and ion flux densities, n_i is the ion density and $\boldsymbol{E}$ is the electric field. Subtracting equation (3) from equation (2) with $n_e = n_i$ gives $\nabla \cdot (\boldsymbol{j}_e - \boldsymbol{j}_i) = 0$. Thus the total particle flux density $\boldsymbol{j} = \boldsymbol{j}_e - \boldsymbol{j}_i$ is conserved as $\nabla \cdot \boldsymbol{j} = 0$. By dividing equation (4) by μ_e, equation (5) by μ_i, and then adding the two equations to eliminate $\boldsymbol{E}$, after substituting $\boldsymbol{j}_e = \boldsymbol{j}_i + \boldsymbol{j}$, we obtain

$$\boldsymbol{j}_i(1/\mu_e + 1/\mu_i) = -(D_e/\mu_e + D_i/\mu_i)\nabla \cdot n_e - \boldsymbol{j}/\mu_e.$$

On dividing the above equation by $(1/\mu_e + 1/\mu_i)$ and substituting this expression for j_i into equation (3), we obtain equation (1), because $\nabla \cdot j = 0$. Equation (1) is much simpler to solve numerically for the distribution of n_e rather than obtaining n_e from the solution of the coupled equations (2)–(5). Furthermore, use of the ambipolar diffusion coefficient is much simpler than solutions involving the difference between the much larger terms for electron drift and diffusion of equation (4). Note that in equation (1) there is an ambipolar diffusion term, but no term in the electric field.

The distribution of the electric field can then be obtained from $\nabla \cdot \boldsymbol{j}_e = 0$, which follows from $\nabla \cdot \boldsymbol{j} = 0$ as we assume $|\boldsymbol{j}_i| \ll |\boldsymbol{j}_e|$. Thus

$$\nabla \cdot (n_e \mu_e \boldsymbol{E} - D_e \nabla n_e) = 0. \tag{6}$$

This current continuity equation is solved in terms of the electric potential, φ, and the electric field is obtained from $\boldsymbol{E} = -\nabla\varphi$. This solution of the current continuity equation differs from conventional solutions in that (a) we use the calculated non-equilibrium electrical conductivity, σ, obtained from the electron density using $\sigma = n_e e \mu_e$, where n_e is the calculated non-equilibrium value of electron density after accounting for diffusion and (b) electron diffusion current is included in addition to the usual electric field component of current. We represent current density by $\boldsymbol{J} = e\boldsymbol{j}$ where e is the electronic charge. Then instead of the usual representation of the current density as dependent only on the electric field $\boldsymbol{E}$, by Ohm's law, $\boldsymbol{J} = \sigma \boldsymbol{E}$, we also include a term, $D_e e \nabla n_e$, to account for diffusion current from electrons so that

$$\boldsymbol{J} = \sigma \boldsymbol{E} - D_e e \nabla n_e. \tag{7}$$

At the anode surface, the equilibrium value of electrical conductivity σ is almost zero, in which case the current would need to be carried by the diffusion term $-D_e e \nabla n_e$. However, in our representation we use a non-equilibrium calculation of n_e to calculate σ and furthermore n_e is not zero at the anode surface if we use the Milne boundary condition.

In figures 1 and 2 we show calculated axial electron densities obtained from such calculations, respectively, for the anode and cathode of a 200 A arc in argon with a tungsten cathode with a 60° conical tip, separated from a water cooled copper anode by 5 mm (Sansonnens *et al* 2000). The arc pressure was 1 bar. The solid curves give the electron density accounting for ambipolar diffusion and the dotted curves show the LTE electron density corresponding to the local temperature, which is shown in the figures by dashed curves. It is seen that for both anode and cathode there are significant departures from the equilibrium electron densities within about 1 mm of the electrode surfaces. Near the anode surface the electron density is greater than the equilibrium value because of the diffusion of electrons from the plasma, which has a high density of electrons. But near the cathode there is a depletion of electrons near the surface. In this case, the temperature and thus the equilibrium electron density is very high, but within a distance of 0.1 mm from the cathode the electrons are depleted due to back diffusion to the cathode. The results of figures 1 and 2 suggest that deviations of charge densities from LTE due to the effects of diffusion are limited to regions within 0.1 mm from the electrodes. The results suggest that we may be able to perform calculations assuming LTE, but simply ignore the non-LTE region in front of the electrodes by setting the mesh size for this region to 0.1 mm.

3. Physical basis of the 'LTE-diffusion approximation' and the diffusion distance D_e/W

In this section we discuss the physical basis of the LTE diffusion approximation in which we use equilibrium values of the electrical conductivity appropriate to the local temperature, without any calculation of non-equilibrium charge densities. The mesh size near the electrode surfaces is chosen to be so coarse as to neglect the detail of non-equilibrium densities near the surface. Our numerical scheme has about 90 nodes in both the radial and axial direction with a non-uniform mesh size. We have the facility to have a fine mesh at the electrode surfaces, both within the plasma region and within the electrode regions, but with a coarser mesh elsewhere. The mesh size is made to vary exponentially with axial position or radius, but preserving the total number of nodes in each of the axial and radial directions at approximately 90. With a step size of 0.1 mm at the electrodes the temperatures adjacent to the electrodes are ~7000 K or more so the electrical conductivity is finite and there are no difficulties with very low values of equilibrium electrical conductivity.

The curves of figures 1 and 2 indicate that the plasma densities are approximately equal to the values for thermodynamic equilibrium, except for a thin region in front of the electrodes. The approximate extent of this region dominated by diffusion can be obtained by examining equation (4) for electron current density, where the axial component, j_{ez}, will be dominant near the anode surface

$$j_{ez} = n_e \mu_e E_z - D_e \frac{\partial}{\partial z} n_e. \tag{8}$$

Within the plasma away from the electrodes, density gradients are small so that the diffusion term in D_e will be small. But at the electrode surface, electron density gradients are extremely

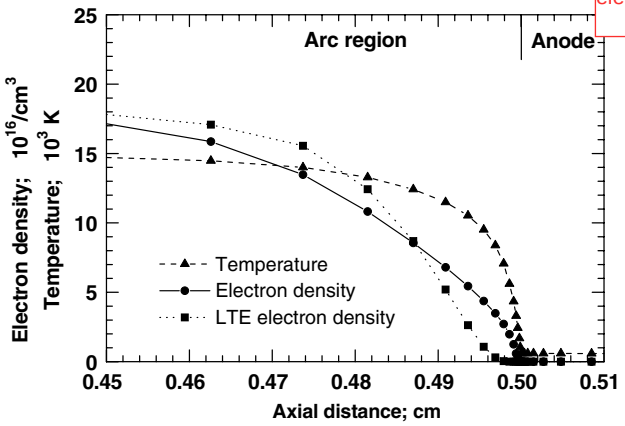


Figure 1. Calculated on-axis non-equilibrium electron densities near the anode of a 200 A arc in argon; Sansonnens *et al* (2000).

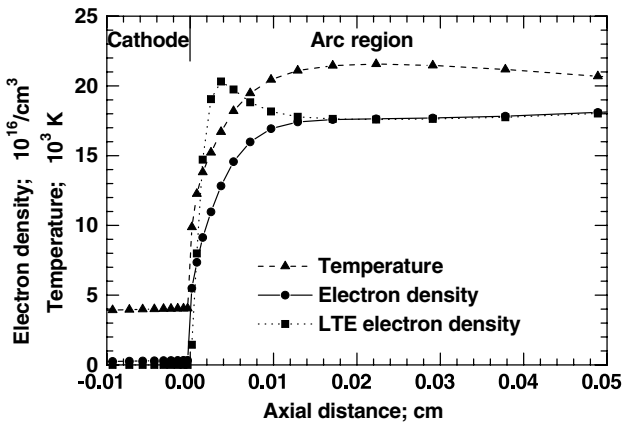


Figure 2. Calculated on-axis non-equilibrium electron densities near the cathode of a 20 A arc in argon; Sansonnens *et al* (2000).

large and the diffusion term will be dominant, certainly for a boundary condition $n_e = 0$ at the anode surface. In our approximate treatment using LTE, there is no account of diffusion so we do not wish to have mesh points so close to the anode surface where the diffusion term of equation (8) is dominant. We obtain a very approximate expression for this 'diffusion length' Λ by equating the two terms of equation (8) to obtain Δz

$$\Delta z = |D_e \Delta n_e/(n_e \mu_e E_z)| = |D_e/(\mu_e E_z)| = |D_e/W| \qquad (9)$$

as the drift velocity $W = \mu_e E_z$. We have set $\Delta n_e \sim n_e$, as we assign a boundary condition of $n_e = 0$ at the anode surface. It is desirable that the mesh size Δz be greater than $\Lambda = |D_e/W|$ so that regions are avoided where diffusion effects are dominant, but for large mesh sizes there will be problems in resolving effects of electrode shape, in particular for the point of the cathode.

Also, from the Einstein relation, $D_e/\mu_e = kT/e$, we obtain

$$\Lambda = (kT/e)/E_z. \qquad (10)$$

When evaluating Λ, we assume properties determined by the arc column for which $T \sim 10\,000$ K, so that $kT/e \sim 1$ V, and the average value of E is 24 V cm^{-1} as the total arc voltage is 12 V and arc length of 0.5 cm. Thus we obtain $\Lambda = 0.04$ cm. In the sheath the value of T will be decreased and values of E

Accelerating field for electrons to counteract back diffusion of electrons from plasma

Retarding force for electrons to maintain same current as in arc column

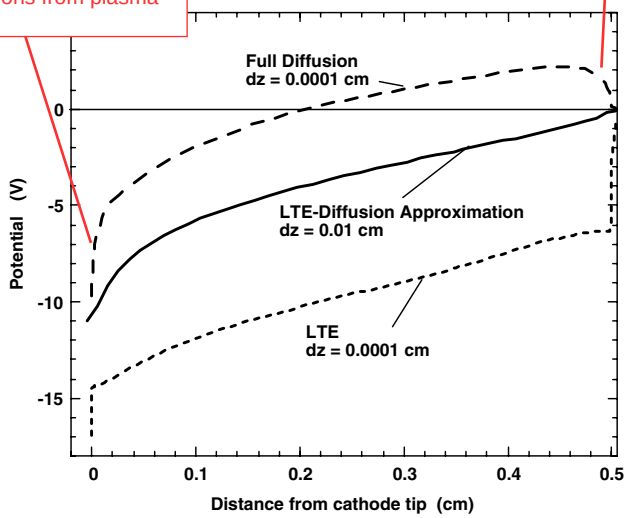


Figure 3. Calculated potential distributions between the cathode, at $z = 0$, and the anode, at $z = 0.5$ cm, for a 200 A arc in argon with a water cooled copper anode. Calculations marked 'Full Diffusion' include ambipolar diffusion of charge whereas those marked 'LTE' and 'LTE Diffusion approximation' omit charge diffusion.

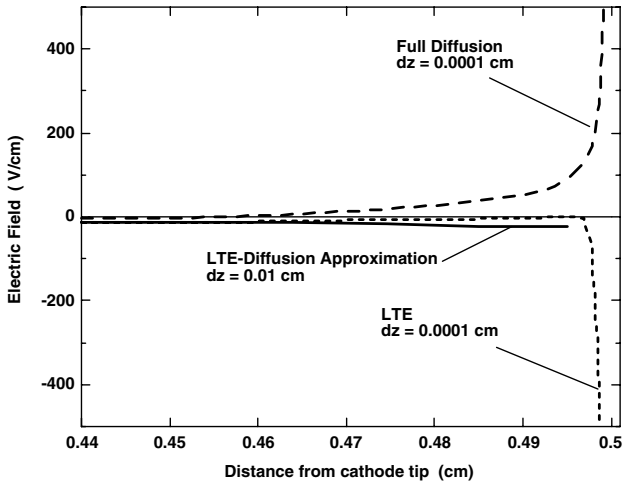


Figure 4. Calculated on-axis electric fields at the anode for the conditions of figure 1.

will be increased so that the theoretical value of Λ is consistent with the value of 0.01 cm from the calculations of figures 3–5, discussed in the next section. The radius of the conical tip of the cathode in the calculations of figures 3 and 4 is 0.01 cm so that choosing a larger mesh size than 0.1 mm reduces resolution near the cathode tip, which influences plasma flow velocities in the arc. In the calculations of figures 10 and 11, also discussed later, we chose a mesh size at the anode of 0.04 cm and a mesh size at the cathode of 0.01 cm, with fair agreement with the much more accurate diffusion calculations.

The value D_e/W also occurs as the characteristic 'diffusion length' for the simple and different case of a very low electron current from a thermionic filament drifting towards an absorbing anode in a uniform electric field. Then, with no ions present, and a current so low that there are no space charge effects, after solving equation (6) with constant drift and diffusion coefficients $W = \mu_e E_z$ and D_e, appropriate to

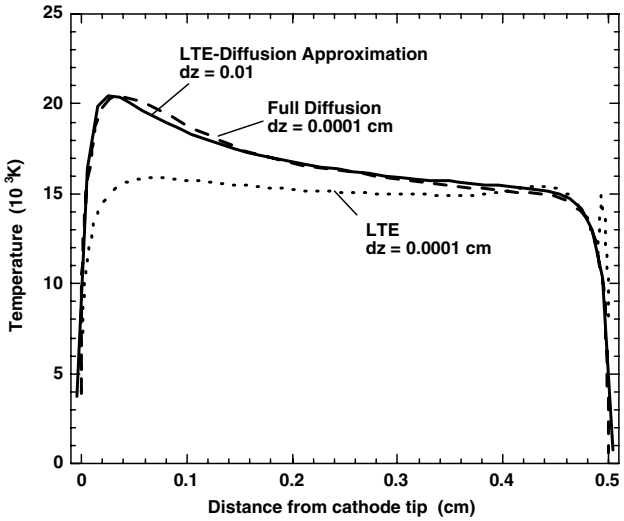


Figure 5. Calculated on-axis temperature distributions for the conditions of figure 1.

the uniform electric field that is applied, we obtain

$$n_e = \frac{j_e}{\mu_e E}\left\{1 - \exp\left[\frac{W}{D_e}(z-h)\right]\right\}$$
$$= n_e(0)\left\{1 - \exp\left[\frac{W}{D_e}(z-h)\right]\right\}, \qquad (11)$$

where j_e, is the electron flux density independent of z. The anode is at $z = h$, at which the electron density is set to be zero as a boundary condition. From equation (11), at a distance from the anode of $|z - h| = D_e/W$, which is the diffusion length, Λ, the electron density has changed from 0 at the anode to the fraction $(1 - 1/e)$ of the value of the electron density far from the anode. So in this case the diffusion length is the distance from the electrode surface over which the electron density changes by the fraction $(1 - 1/e)$.

The quantity D_e/W is also a significant quantity in electron transport theory, being the energy relaxation distance over which electrons come to equilibrium energy in a uniform electric field (Parker and Lowke 1969).

4. Effect of neglecting diffusion currents

Figures 3–5 show calculated on-axis distributions of potential, electric field and temperature, respectively, between the electrodes of a 200 A arc in argon with a cathode having a 60° conical tip and a water-cooled copper anode. Each figure shows distributions for three calculations: (*a*) dashed curves, which include an accurate account of electron diffusion to the electrodes, with a mesh size at the electrodes of 0.0001 cm, which we refer to as the 'full diffusion' calculation. (*b*) Solid curves, which omit diffusion and assume LTE, with a mesh size at the electrodes of 0.01 cm, which we refer to as the 'LTE-diffusion approximation'. (*c*) Dotted curves, obtained from a similar LTE calculation to (*b*), but with a mesh size at the electrodes of 0.0001 cm. The results of the LTE calculation (*c*) become unrealistic because the mesh size is less than the diffusion length and so the electrical conductivities are set to the very low equilibrium electrical conductivities corresponding to the electrode temperatures.

The character of each of these three calculations is quite different. A feature of the 'Full Diffusion' calculations in the figures, is the presence of a potential maximum at the anode (see figure 3). This maximum arises because the diffusion current of electrons to the anode is so large that a retarding field is necessary at the anode to maintain the same electron current as occurs in the arc column. This strong retarding field is seen in the 'full diffusion' results of figure 4. There is a similar change in voltage at the cathode, to provide an increased accelerating field at the cathode, to counteract the strong electron current by back diffusion of electrons from the plasma to the cathode. The 'Full Diffusion' curve of figure 3 is consistent with the concepts of conventional probe theory (Cobine 1958) where the plasma near a metal probe attains a potential approximately kT/e higher than that of the probe, as electrons escape from the plasma to the probe until the average electron energy, kT/e, is balanced by the retarding potential of the plasma. For the arc plasmas the temperature is approximately 20 000 K, so that kT/e is about 2 V. The LTE-diffusion calculations, with a mesh size of 0.01 cm, effectively assume that the temperature of the plasma at the electrode surface is the same as the temperature of the plasma at the first mesh point away from the electrodes. These calculations cannot account for or show the maximum in the potential at the anode resulting from electron diffusion, or the large retarding field at the anode, shown in figures 3 and 4.

The increase in electron current due to diffusion at the anode and the decrease in electron current due to the retarding field, are approximately self-cancelling. Thus, omission of diffusion currents and the retarding electric field in the vicinity of the electrodes is found to be a good approximation. Furthermore, from figure 3 it is seen that the derived total arc voltage of 11 V obtained from the LTE-diffusion approximation is close to the total arc voltage of 10 V obtained from the full diffusion calculation. This approximate equality arises because the further largely self-cancelling effect that the lowering of the cathode voltage due to diffusion effects largely balances the change in the anode voltage, also due to diffusion effects. Thus the total enthalpies delivered to the anode are equal in the two calculations to within 10%, as also shown in figure 3. Furthermore, the derived temperature profiles for the two calculations are also very similar, as seen in figure 5.

LTE calculations for very small mesh sizes become increasingly unrealistic, as is seen in figures 3–5 for the curves marked LTE with a mesh size of 0.0001 cm. High fields are required to make possible conduction of the total current through the cold region in front of the electrodes where the electrical conductivity is very low. The calculated arc voltage then becomes 17.5 V (see figure 3) which is unrealistically high. In figure 5, it is seen that this high field has caused the formation of a high temperature plasma, or arc spot, immediately in front of the anode. These calculated results and the predicted nature of the discharge vary strongly with step size because of the differing effective electrical conductivities in front of the electrodes.

Table 1 shows calculations of total arc voltage, which is closely related to the total heat transfer to the anode, and also the calculated maximum temperature at the anode, for an arc of 200 A, with a water-cooled copper electrode. Results are given

Table 1. Effect of mesh size at electrodes. Comparison of calculations including diffusion of charges to electrodes with calculations assuming LTE charge densities. 200 A, argon, copper anode. Values in parenthesis are calculated with the Milne boundary condition at the anode.

Anode mesh (cm)	Cathode mesh (cm)	Voltage (V)		Anode temp (K)	
		Full Diffusion	LTE	Full Diffusion	LTE
0.000 01	0.000 01	10.7 (10.3)	No solution	642 (601)	No solution
0.000 05	0.000 05	10.5 (10.3)	22	620 (595)	>20 000
0.000 1	0.000 1	10.3 (10.2)	17.5	606 (586)	1 470
0.001	0.001	9.9 (9.9)	14.7	573 (570)	1 140
0.01	0.01	9.9	11.4	570	750
0.02	0.01	10.1	10.9	563	684
0.04	0.01	10.3	10.6	541	628

for widely different mesh sizes near the electrodes, (*a*) for calculations, marked 'Full diffusion', with a full account of the electron density by diffusion in the electrode regions and (*b*) for calculations marked LTE, which omit diffusion. The calculations in brackets for the diffusion calculations are for a 'Milne' boundary condition for electron density at the anode, discussed more fully in section 6. The mesh is non-uniform and is chosen to vary exponentially with position. It is seen that (1) the LTE results become increasingly unrealistic for small mesh sizes (2) the results including an account of electron diffusion are remarkably independent for mesh sizes varying by a factor of over 1000, particularly if one takes the results for the Milne boundary condition for small mesh sizes and (3) the predictions for total arc voltage and maximum anode temperature for mesh sizes of ~0.01 to 0.04 cm at LTE agree well with the predicted results including electron diffusion for very small mesh sizes of 0.00001 cm. This very small mesh was used to make sure that recombination coefficients are accurately represented, as is discussed in section 6. We refer to the LTE results, with a mesh size at the electrodes of 0.01–0.04 cm, as those of the 'LTE-Diffusion' approximation.

5. Boundary conditions for heat transfer at the electrodes

The dominant boundary condition for heat transfer at the electrodes involves $j\varphi$, where φ is the work function of the electrode. At the anode, $j\varphi$ represents the heating process of electrons 'condensing' in the anode and at the cathode $j\varphi$ represents a cooling process due to the energy required for the liberation of electrons from the cathode. The remaining principal heat transfer from the plasma to the electrodes is represented by $-\kappa\nabla T$, where κ is the equilibrium thermal conductivity of the gas at the temperature near the electrode surface. We also include a term for the emission of radiation from the surface of the electrode, although for copper electrodes this term is negligible as the surface temperatures of copper are low due to the high thermal conductivity of copper and also due to the low melting and boiling temperatures of copper. At the cathode we have an additional boundary condition of ion heating given by $j_i V_i$, where the ion flux density, j_i, at the electrode surface, is assumed to be given by the difference between the electron flux density j as obtained from equation (7), and flux density from thermionic emission j_R, given by $j_R = AT^2 \exp[-\varphi_e e/(kT)]$; V_i is the ionization potential of the gas, φ_e and A are the constants for thermionic emission of the cathode. For thermionic emission we have taken $\varphi_e = 2.63$ eV, the work function of thoria. However, for thermionic cooling of the cathode we have used the tungsten work function of 4.5 eV, as tungsten is the main material of the cathode and there is an additional energy transfer when electrons are conducted from tungsten to thoria (Sansonnens *et al* 2000).

In many other investigations, several additional terms have been used to account for other processes that result in heat transfer at the anode. For example, Lancaster (1986) and Chen and Li (2001) use a heating term jV_A at the anode where V_A is the anode fall, usually regarded as necessary to produce positive ions. In our analysis, the positive ions originate from the thermally ionized plasma at the anode. The anode 'fall' of 2.3 V is in fact negative, as in figure 3, due to the necessity of producing a retarding field immediately in front of the anode to reduce the diffusion current to the anode to that of the main arc column. These effects are fully accounted for in the diffusion calculations without need for the extra term jV_A, which is also omitted in the LTE calculations. A term involving jkT/e is also frequently introduced to account for energy transfer of electrons at high temperature to the anode (Amakawa *et al* 1998, Lago *et al* 2004). In our calculations it is assumed that temperatures of all particles are equal and that electron temperatures on entering the anode equal the anode temperature, so that this term does not apply. Such energy transfer is accounted for by the thermal conductivity, κ, which is calculated assuming LTE and includes the electron component. As kT/e is of the order of 1 V and the total arc voltage is ~10 V, neglect of this term would result at most in a 10% underestimation of the total energy delivered to the anode.

In chen and Li model, this fall is not modeled by their equations and therfore, additional term is required

6. Boundary conditions for the electron density at the anode

Since all electrons impinging on the anode are absorbed but no electrons are emitted or reflected by the anode to the plasma, the correct boundary condition at the anode is not for the electron density to be zero but that the gradient of the electron density at the anode is set so that the extrapolated density is zero at about one mean free path beyond the anode; see for example Lindenfeld and Shizgal (1983). The mean free path varies significantly with electron energy because of the varying electron–atom cross section for argon with electron energy. Furthermore, the mean free path is inversely proportional to gas density, which in turn depends on gas temperature. For the Milne boundary condition we used a simple approximation for the extrapolation length of 0.004 cm, for all radii at the anode.

In figure 6 we show calculations of electron density on axis at the anode for a 200 A arc in argon, the results being obtained from a solution of equation (1). The results at each mesh point are shown, indicating our non-uniform grid. As in

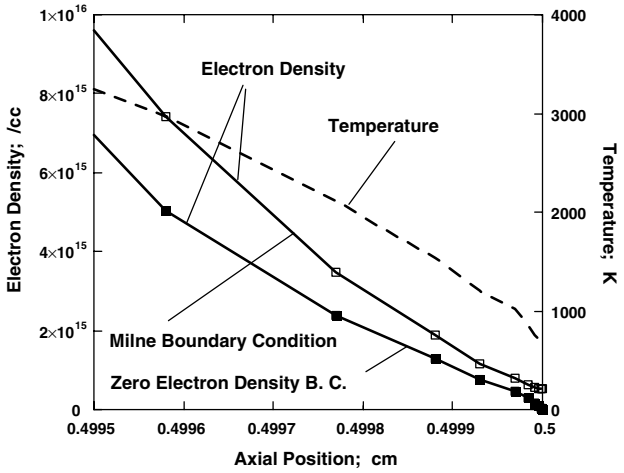


Figure 6. Calculated electron densities and temperatures in the anode sheath region for a 200 A arc showing differences in the boundary conditions at the anode of zero electron density and the Milne boundary condition, where the boundary condition is such that the extrapolated electron density is zero at one mean free path λ beyond the anode surface.

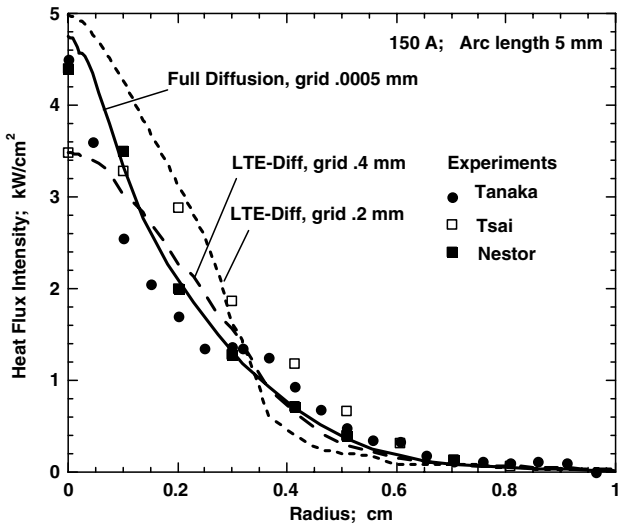


Figure 7. Comparison of calculated heat flux densities at a copper anode for a 150 A arc. The 'LTE-Diffusion approximation' and the more accurate full diffusion calculation are in close agreement. Experimental values are shown as points.

the computation of Sansonnens *et al* (2000), we have used the recombination coefficients of Hoffert and Lien (1967) at low temperatures, which are proportional to $1/T^{4.5}$. The derived temperature within the sheath varies very rapidly with position, as also shown in figure 6. It is necessary to use very small mesh sizes in the sheath to resolve the variation of the recombination coefficient with position, as is required for the solution of equation (1). There will be conditions of temperature non-equilibrium within the sheath (Tanaka and Ushio 1999), but we were nevertheless interested to complete a solution of the equations assuming that equilibrium temperatures exist, to obtain values of heat flux density and current density as a function of radius. These values are presented in figures 7–10, discussed in the next section.

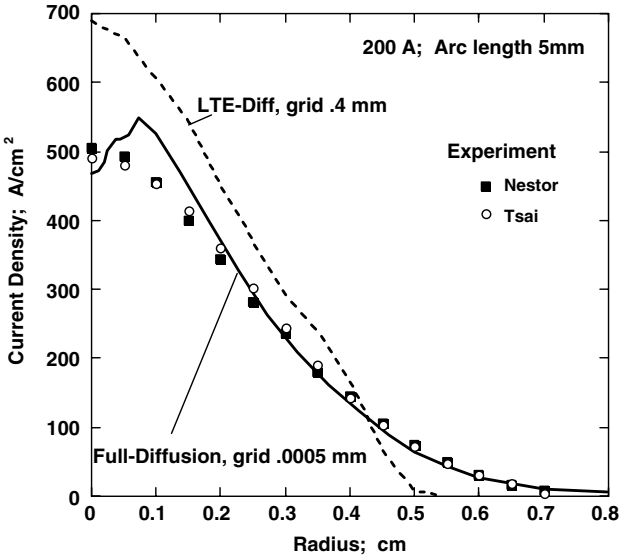


Figure 8. Comparison of calculated current densities at a copper anode for a 200 A arc. The 'LTE-Diffusion approximation' and the full diffusion calculation are in approximate agreement. Experimental values are shown as points.

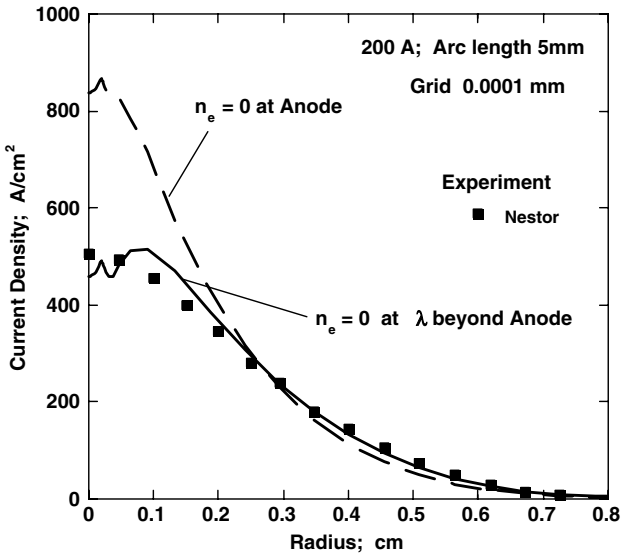


Figure 9. Comparison of the calculated current densities at the anode for a 200 A arc from a full diffusion calculation, but using the two different boundary conditions for the electron density at the anode of figure 6.

7. Comparison of predictions of heat and current density with experimental results

Figure 7 shows a comparison of predictions of the heat flux density to the anode, shown by curves, with experimental results, shown by points, for a 150 A arc in argon with a 60° thoriated tungsten cathode separated from the anode by 5 mm. The experimental method for all three experimental investigations used the method initiated by Nestor (1962) where measurements are made by calorimetry of the heat delivered to each half of a divided anode for various positions of the vertical cathode moved laterally relative to the cathode.

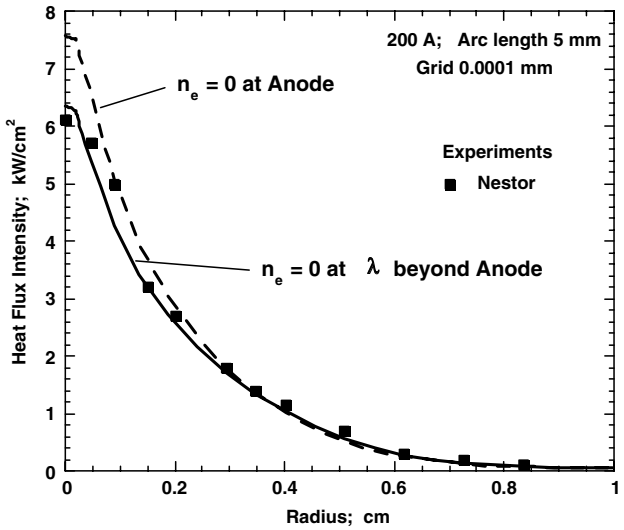


Figure 10. Comparison of the calculated heat flux density at the anode for a 200 A arc from a full diffusion calculation, but using the two different boundary conditions for the electron density at the anode of figure 6.

Results of heat flux density as a function of radius have to be obtained by Abel inversion, so that it is particularly difficult to obtain the heat flux density at the centre of the arc, as this estimate needs to be obtained by obtaining differences between measurements of the integrated heat energy received from all other arc radii contributing to the total heat transfer to the electrode segment. The experimental procedure is also described in the more recent paper of Tanaka *et al* (2002). Results of Tsai and Eagar (1985), also shown for current density in figure 7 are fitted to conform with the function $\exp(-r^2/2s^2)$ where s is a fitting parameter. The experimental points marked Tanaka in figure 7 are from Ushio *et al* (2004).

The theoretical curves have two input components to the heat flux density: (1) the contribution $j\phi$, from the absorption of electrons and (2) heat conduction from the arc to the anode. Contributions from both the emission of radiation from the anode and the absorption of arc radiation from the arc by the anode are small. The calculated solid curve represents mesh sizes of both 0.000 05 cm and 0.000 01 cm, which are in agreement and also are in agreement with the experimental results, shown by points. The calculations for the LTE-diffusion approximation, marked 'LTE' in the figure, are in fair agreement with the experimental results and the full diffusion calculation, particularly for a mesh size at the anode of 0.04 cm. The results for the LTE-diffusion approximation are sensitive to the mesh size, increasing by 30% at the axis when the mesh size at the anode is halved to 0.02 cm.

Similar results are shown in figure 8 for current density as a function of radius. Again the full diffusion calculation is in good agreement with the experimental results. The theoretical curve has an off-centre maximum, which occurs because there are two components to the total electron current density, firstly, due to the electric field and secondly, due to the diffusion current. It is noted that quite a number of the experimental results of Nestor (1962) show off-centre maxima for both the current density and heat flux density distributions. As in the case of heat flux density, curves for the LTE-diffusion approximation for a mesh size of 0.04 cm are in acceptable agreement with experimental results. The full diffusion results were calculated using the Milne boundary condition. Agreement between theoretical predictions and experimental results for current density is not as good as for heat flux density because the current density depends on the difference between an electric field component and a diffusion component. The diffusion component depends on the gradient in the electron density and is more difficult to calculate accurately.

Figure 9 shows results obtained with the two different boundary conditions for the current density at the anode. The two curves differ markedly, and it is the curve with the Milne boundary condition that is in agreement with the experimental results of Nestor. The curve of this figure was calculated for a mesh at the anode of 0.000 01 cm and is in good agreement with the curve of figure 8, calculated with a mesh at the anode of 0.000 05 cm. Curves calculated with the $n_e = 0$ condition at the anode actually diverge for small mesh sizes as indicated by the results in table 1. Such a dependence of results on small mesh size was also obtained in the investigation of Sansonnens *et al* (2000) as indicated in their figure 10. This divergence is explained because fixed boundary conditions for electron density defines a solution of n_e from equation (1) which is independent of total current. The Milne gradient boundary condition enables stable solutions to be obtained at very small mesh sizes.

Figure 10 shows results similar to those of figure 9, but for the heat flux density. Unlike the curves of figure 9, the heat flux curves differ only slightly for the two boundary conditions of electron density at the anode. The results obtained using the Milne boundary condition give slightly better agreement with experimental results.

Figures 11 and 12 show a comparison between the two methods for a calculation of anode melting. The anode in this case is of stainless steel with the volume at temperatures above the 1750 K contour being molten. Temperature profiles are calculated for a 150 A arc, including a calculation for the weld pool and convective flow in the weld pool. The mesh size at the cathode was 0.01 cm and at the anode 0.04 cm. The left-hand figure in figures 11 and 12 shows the calculation with the full diffusion calculation whereas the right-hand figure shows the LTE-diffusion approximation. It is seen that there is a negligible difference between the two calculations for the volume of molten metal or 'weld profile'. Similarly, calculations of figure 12 show arc and plasma flow. Again close agreement is obtained.

Agreement in figures 11 and 12 in predictions of weld pool shape between the 'LTE-diffusion' approximation and the more exact diffusion calculation is facilitated by two differences from the previous calculations. (1) The calculations are for conditions after 20 s in a time-dependent calculation so that comparisons benefit from any effect on the results of the initial temperature of the metal, which was taken to be at ambient in each case. (2) The effect of radial convection in the molten weld pool tends to remove the effects of differences between the two calculations arising from the radial variations in current and heat flux density shown in figures 4 and 5.

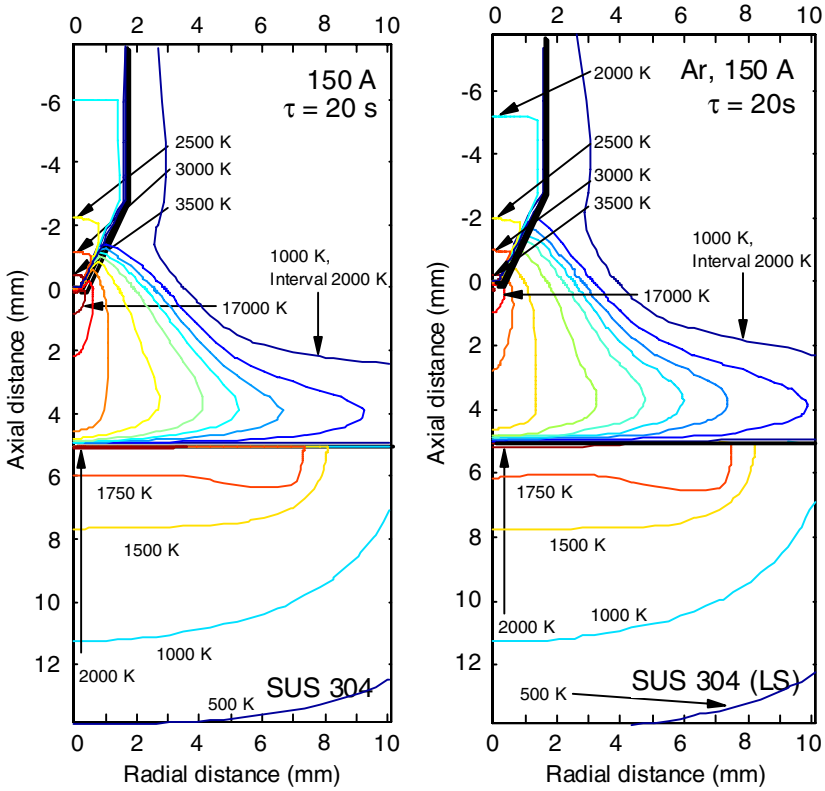


Figure 11. Calculated arc and electrode temperatures after 20 s for 150 A and a stainless steel work-piece. The left-hand half of the figure is for a full account of electron diffusion and the right hand half is for the LTE-diffusion approximation.

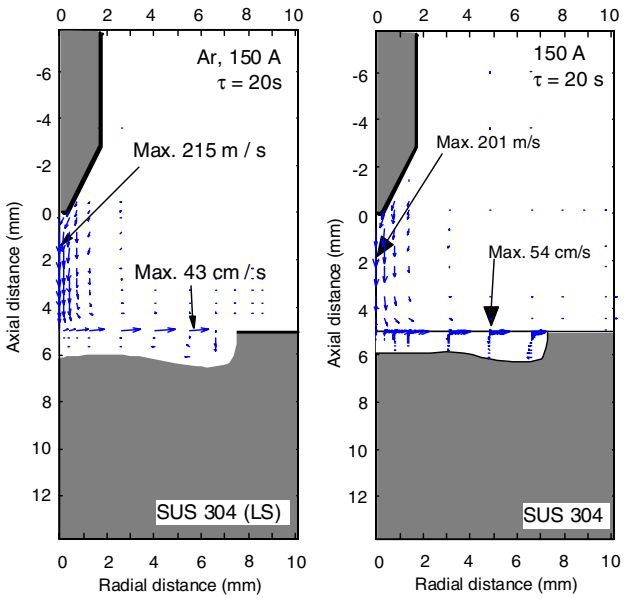


Figure 12. As for figure 11, but showing calculations for convective flow.

8. Conclusion

From a comparison of calculations of arc properties derived from a unified electrode-arc treatment with account taken of electron diffusion, it is concluded that reasonably accurate calculations of arc properties can be made assuming LTE throughout the region and omitting diffusion effects, provided that mesh sizes at the electrodes are ~0.01–0.04 cm. This distance is approximately the 'diffusion length' of electrons in the arc plasma, defined by $D_e/W \sim (kT/e)/E$ for the arc plasma. We call the use of LTE with this mesh size at the electrodes the 'LTE-diffusion approximation'. Use of this approximation, for a 5 mm long arc of 200 A, with a thermionic cathode, gives predictions of total arc voltage, electrode temperatures, arc temperatures and radial profiles of heat flux density and current density at the anode. These predictions are in approximate agreement with more accurate calculations which include an account of the diffusion of electric charges to the electrodes and also with experimental results. The effects of electron diffusion currents to the electrodes, which cause large changes in the local electric fields, are not simulated in the approximation either in magnitude or even in sign. These effects can be omitted in the approximation because total heat transfer to the electrodes is dominated by the effects in the arc column away from the electrodes and also because the effects at the two electrodes are to some extent self-cancelling.

Calculations accounting for diffusion of charges to the electrodes are in good agreement with experimental results of current density and heat flux density as a function of radius if the Milne boundary condition is used at the anode, rather than the boundary condition that charge densities are zero at the anode. The Milne boundary condition imposes a gradient at the anode surface to make the extrapolated charge density zero at about one mean free path beyond the surface.

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